Trapped by Cash:

A Theory of Voluntary Informality Among the Economically Non-Viable*

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Abstract

Why do loss-making informal firms persist? Standard models answer with rationing: formal employment is scarce, and agents who prefer formal work cannot find it. This paper proposes a different answer for a large fraction of the loss-making informal population. We develop a model in which informal self-employment generates two forms of embedded capital: productive capital, which raises economic output, and relational capital—loyal customer relationships, supplier credit, deferred-rent arrangements, location tenure—which raises cash flow and absorbs idiosyncratic shocks without raising economic income. Solving the agent's dynamic programming problem yields a trap condition, $CE(CF) \geq w^*$, under which agents decline formal offers even when freely available. The condition is independent of both the offer arrival rate and the

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job separation rate—the two parameters that active labour market policy targets. Crossing this condition with the viability condition $\pi^{\mathrm{adj}} \gtrless 0$ generates a four-type equilibrium partition of the informal sector as a theoretical result, not an empirical classification. The two loss-making types—self-trapped (Type C) and rationing-constrained (Type D)—require entirely different policy responses. Standard active labour market policy is correctly targeted at Type D and irrelevant for Type C. We validate the model using seven-day double-entry financial diaries for 500 informal firms across six sectors and nine departments of urban Bolivia. The diary instrument, which records every transaction at daily frequency, reveals that between 59 and 75 percent of loss-making firms satisfy the trap condition across all defensible values of the formal wage threshold. A sign-reversal test—family hours raise cash flow and reduce adjusted income simultaneously for loss-making firms—provides direct evidence that agents optimise over cash flow rather than economic income, confirming the model's selection condition. Because relational capital accumulates with operation, the model further predicts the trap is an absorbing state that firms age into: consistent with this, the trapped share of loss-making firms rises monotonically with enterprise tenure, from 62 percent among entrants to 78 percent among firms operating sixteen years or more.

JEL: O17, J24, J46, D21, D91.

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1 Introduction

Bolivia has the largest informal economy in Latin America and, by IMF estimates, the largest in the world: 62.3 percent of GDP, the highest share among 158 countries surveyed by Schneider (2018). Eighty-five percent of Bolivian workers are informally employed (ILO, 2024)—a figure that has proved stubbornly resistant to decades of active labour market programmes, formalisation incentives, and training initiatives. Bolivia is therefore the right place to study informal persistence: not an outlier, but the extreme case of a phenomenon common across the developing world.

Active labour market programmes targeting the informal sector rest on a maintained hypothesis: loss-making informal workers prefer formal employment and are prevented from obtaining it by the scarcity of formal positions. Under this hypothesis, the policy prescription is direct— raise the probability of formal offers, through job matching services, hiring subsidies, or training programmes. The evidence on programme effectiveness is, to put it charitably, mixed (Card et al., 2010; Betcherman et al., 2004). This paper argues the maintained hypothesis is wrong for a majority of the loss-making informal population, and that the policy failure follows directly from the theoretical error.

The theoretical error is one of selection, and a concrete example makes it vivid. Consider an owner-operated food stall—a taquero— who has worked the same street corner for a decade. His neighbours have become customers; they pay yesterday's price and expect tomorrow's tab. His supplier extends thirty-day credit; a rival vendor on a new corner would pay cash. His landlord has not raised rent in three years; the going rate is twice what he pays. Now suppose a surveyor asks: how much did you earn last month? The taquero reports cash in minus cash out—his *conventional profit*—which is healthy, well above the minimum wage. Suppose further that a development economist, equipped with this survey response, classifies the taquero as "voluntarily informal" and leaves him alone. The classification is wrong. The taquero's conventional profit is healthy precisely because his relational capital—the loyal customers, the supplier credit, the below-market lease—is converting inputs into cash flow at a rate that obscures the underlying economics. Price all three relationships at market rates and his income turns negative. He earns below his own market wage; his firm is economically non-viable. But he cannot leave. Every formal offer he receives would extinguish the relational capital that generates his cash flow. He declines not because formal employment is scarce, but because informal operation—through accumulated relationships—dominates formal employment on the only metric that governs his budget constraint: cash.

This paper builds a model around that mechanism and tests it against transaction-level data.

Standard models of informal sector persistence (Fields, 2011; Harris and Todaro, 1970; Maloney, 2004) characterise informal self-employment as a residual state: agents in it either prefer formal employment and are rationed out, or prefer informality because their informal earnings exceed the formal wage. The first group is the target of active labour market policy. The second group—voluntarily informal—is left alone, since there is no market failure to correct. The classification is clean in theory. In practice, it has a critical flaw: it treats the comparison between informal earnings and the formal wage as observable from standard survey data, when in fact the relevant comparison is between informal *cash flow* and the formal wage— and cash flow is not observable from surveys.

The distinction between cash flow and economic income in informal firms is not a measurement detail. It is the central economic mechanism of the paper. Owner-operated informal firms incur costs that are economically real but generate no cash outflow: the opportunity cost of the owner's time, economic depreciation of equipment, the implicit cost of operating below market rent through long-standing landlord relationships. An informal firm can simultaneously have negative economic income—the owner earns below their market wage, capital earns below its opportunity cost—and strongly positive cash flow. For an agent comparing informal operation against formal wage employment, what enters the comparison is cash flow, not economic income. The budget constraint runs through cash, not accounting profit.

This observation has a model, and the model has sharp implications. We develop a three-state dynamic programming model in which agents choose between informal self-employment, formal employment, and unemployment. Informal self-employment generates two distinct forms of embedded capital that accumulate through operation: *productive capital* (e^P), which is labour-augmenting knowledge that raises economic output, and *relational capital* (e^X), which is the social infrastructure of informal operation—loyal customers, supplier credit, deferred-rent arrangements, location rights—that raises cash flow and absorbs productivity shocks without raising economic income.

The two forms of capital have identical laws of motion: both accumulate with tenure in informal self-employment and depreciate upon exit. This makes them observationally indistinguishable from any survey instrument that records tenure. But their economic functions are sharply different: productive capital determines whether the firm is viable ($\pi^{\text{adj}} \gtrless 0$), while relational capital determines whether the agent would accept a formal offer if one arrived ($CE(CF) \gtrless w^*$).

Solving the Bellman equations yields the paper's central theoretical result.

> *An agent in informal self-employment declines a formal offer if and only if the certainty equivalent of informal cash flow exceeds the formal wage:*

$CE(CF) \geq w^*$. This condition is independent of the formal offer arrival rate ρ and the job separation rate λ.

The independence from ρ and λ is not a technicality. It is the policy result. ρ is exactly what active labour market policy raises: more matching services, more intermediation, more hiring subsidies. λ is exactly what employment protection legislation targets: reduced separation reduces the cost of formal employment from the worker's perspective. Both instruments are irrelevant for agents satisfying the trap condition. An agent who would decline every formal offer derives no benefit from receiving more of them. An agent for whom the informal sector dominates regardless of job security is unaffected by employment protection.

Crossing the trap condition ($CE(CF) \gtreqless w^*$) with the viability condition ($\pi^{\text{adj}} \gtreqless 0$) yields a four-type equilibrium partition that follows as a corollary of the model. The two loss-making types are what matter for policy: Type C agents ($\pi^{\text{adj}} \leq 0$, $CE(CF) \geq w^*$) are economically non-viable but voluntarily informal— their relational capital sustains cash flow above the formal wage. Type D agents ($\pi^{\text{adj}} \leq 0$, $CE(CF) < w^*$) are economically non-viable and rationing-constrained— they would accept a formal offer and are prevented only by $\rho < 1$. Standard active labour market policy correctly targets Type D. It is irrelevant for Type C. The two populations are observationally identical on standard survey variables: age, tenure, and hours worked do not separate them because both forms of embedded capital accumulate through the same law of motion.

Empirical validation. We test the model using a purpose-built measurement instrument: a seven-day double-entry financial diary administered to 500 informal firms across six sectors and nine departments of urban Bolivia in September–October 2024. The diary records every transaction at daily frequency, distinguishing cash and non-cash flows with the precision required to separately compute conventional profit, adjusted net income, and operating cash flow for every firm. This level of granularity is unavailable in any standard household survey; it is what makes the trap condition testable rather than merely theoretical.

The empirical strategy proceeds in three steps, corresponding to three claims the model makes.

The first claim is accounting: the gap between cash flow and adjusted income is driven entirely by non-cash costs—imputed owner labour and depreciation. The diary verifies this at 100 percent precision across all 500 firms. No cash changes hands to produce the economic loss; loss-making firms are financially solvent every day.

The second claim is the central quantitative result: between 59 and 75 percent of loss-making firms satisfy $CF \geq w^*$ across all defensible values of w^* in the empirically plausible range. The share is stable—it does not collapse as w^* rises within the range. At the statutory minimum wage (Bs 500 per week), the trapped share is 75 percent. At the shadow wage opportunity cost (Bs 611 per week), it is 62 percent. The implication is immediate: standard active labour market policy, which treats all loss-making informal firms as rationing-constrained, misallocates resources on a large scale.

The third claim is the mechanism test. If agents optimise over cash flow rather than adjusted income—as the trap condition requires—then inputs that raise cash flow without raising adjusted income should deepen the trap. Family labour hours are exactly such an input: they substitute for hired labour, reducing the cash wage bill and raising CF, while adding imputed labour cost and reducing π^{adj}. For loss-making firms, the coefficient on family hours in the adjusted income equation is -7.6^{***}; in the cash flow equation for the same 214 firms it is $+5.4^{***}$. The sign reversal is not a statistical curiosity. It is direct evidence that agents optimise over CF, consistent with the model's selection condition and inconsistent with any model in which agents optimise over economic income.

Relation to the literature. The evidence in this paper rests on measurement that existing datasets do not provide. Every major cross-country dataset on informality—SEDLAC, LSMS, I2D2—records self-employment income with a single survey question: "How much did you earn from your business last month?" That question elicits conventional profit at best. It cannot recover adjusted income because it does not separately observe owner hours at a market wage, economic depreciation, or the gap Δ that is the paper's central object.

The consequences are not small. The dual-sample design in this paper—500 firms observed simultaneously by diary and by the same instrument a household survey would use—quantifies the damage directly. The survey over-counts loss-making firms by 64 percent (351 vs. the diary's 214), misclassifies 109 genuinely viable firms as trapped, and fails to identify 63 of 150 true Type C firms. Survey precision for identifying trapped loss-making operators is 41 percent: false positives outnumber true positives. Any cross-country comparison built on survey income would not replicate the mechanism—it would replicate the pattern of measurement errors. The mechanism this paper describes has not been documented in the cross-country literature because the data required to document it have not existed. This paper connects five literatures.

Informal sector structure. The segmented labour market tradition (Harris and Todaro, 1970; Todaro, 1969) views informal employment as involuntary—a residual absorbing workers rationed out of formal employment. The competitive tradition (Maloney, 2004; Loayza, 2007) views it as largely voluntary—agents choosing lower taxes and greater flexibility. Ulyssea (2018) provides the most careful structural treatment, showing both types coexist in Brazil. The present paper sharpens the voluntary/involuntary distinction in a direction the existing literature has not considered: voluntariness among loss-making informal firms—the population every previous paper treats as involuntary—is not only possible but predominant in our data. The distinction rests on cash flow rather than on preferences elicited by survey questions about job preferences, which are subject to anchoring biases that make them unreliable for policy targeting. Poschke (2013) shows that subsistence and opportunity entrepreneurs differ in their entry conditions; the present paper shows they also differ in the mechanism that keeps them in the informal sector—and that this mechanism is invisible to the entry-condition approach.

Misallocation and aggregate productivity. Hsieh and Klenow (2009) establish that misallocation of capital and labour across heterogeneous producers accounts for a large share of the TFP gap between developing and developed economies. Bento and Restuccia (2017) extend this to show that informality amplifies misallocation through size-distorting regulations. La Porta and Shleifer (2014) document that the informal sector in developing countries is dominated by subsistence firms that remain small and unproductive indefinitely—a dual economy that does not converge. The present paper provides a microfoundation for this persistence. Type C firms destroy economic value ($\pi^{\text{adj}} \leq 0$) but remain because their relational capital sustains cash flow above the formal wage. They are not growing toward viability and not exiting toward formal employment. They are the microeconomic unit that sustains the dual economy: permanently non-viable, financially solvent, and rationally stationary.

Capital returns in microenterprises. de Mel et al. (2008), McKenzie and Woodruff (2008), and Meager (2019) document high and heterogeneous returns to capital among informal microenterprises. The standard interpretation is heterogeneous ability. The present paper offers a different decomposition: for Type C firms, capital raises revenue without proportionally raising cash flow, because the relational structure—fixed-price supplier arrangements, loyalty obligations, location-based pricing—absorbs the output gain before it reaches cash. The pooled capital-revenue elasticity overstates the production-channel return by 25 percent and overstates the welfare-relevant cash flow return by a factor of 3.5. A capital transfer programme evaluated using pooled returns will overstate its benefit to the target population by this factor.

Networks and mobility traps. The closest mechanism in the existing literature is Munshi and Rosenzweig (2016), who show that Indian caste-based insurance networks rationally suppress rural–urban migration: the network's services are lost upon exit, so workers decline a large urban wage premium. The present paper shares the architecture—non-portable relational capital sustaining an occupational state despite an apparent income gap—but differs in three respects. The margin is occupational rather than spatial; the sustaining service is cash flow rather than insurance, which makes the mechanism operate through the *observability* of income concepts (agents see CF, not π^{adj}) rather than through risk sharing; and where Munshi and Rosenzweig (2016) infer the network's value structurally, the diary measures the relational cash channel directly, transaction by transaction. Breza et al. (2021) document a complementary friction—rationing in casual labour markets—that our Type D population faces by construction; the contribution here is to show that a majority of loss-making informal firms are not in that rationed state at all.

Entrepreneurship and occupational choice. Gollin (2002) shows that adjusting for self-employment income substantially raises measured labour shares in developing countries, implying that conventional national accounts misattribute self-employment income to capital. The present paper operates at the firm level and shows the same misattribution is endemic within the informal sector itself: conventional profit mismeasures the income of informal self-employed by omitting the opportunity cost of owner time, creating the appearance of positive returns where economic returns are negative. Hurst and Pugsley (2011) show that most small business owners in the United States earn less than they would as wage workers once owner time is correctly priced. Our results extend this finding to the informal sector of a developing economy and show it has first-order implications for occupational choice dynamics that are absent in the US context, where relational capital is not the primary mechanism sustaining below-market returns.

Financial diaries and household finance. Collins et al. (2009) introduce the diary methodology to document the financial lives of the poor across three countries. Dupas and Robinson (2013) use diary-adjacent mechanisms to document savings constraints in Kenya. de Mel et al. (2009) provide the single most important measurement result for the present theory: the correlation between self-reported profits and calculated revenue-minus-expenses is only 0.2–0.3 among microenterprises, and 30 percent of firms report positive profits while having negative calculated profits. This divergence is a direct empirical analog of the cash flow veil—but no prior paper has linked this measurement fact to exit decisions or equilibrium firm persistence. Drexler et al. (2014) demonstrate the behavioral foundation: standard accounting training for micro-entrepreneurs had no significant effect on business practices, only simplified rule-of-thumb training improved outcomes. If owners cannot even separate business and household accounts, they cannot compute depreciation or the opportunity cost of their labour. The present paper uses the diary instrument for a different purpose: not to document the level of financial flows but to decompose them into economic income and relational cash contributions, making the trap condition directly observable at the transaction level. The double-entry structure of our diary—which distinguishes cash flows from accruals and records both sides of every transaction—goes beyond existing diary instruments and is the methodological contribution that makes the empirical tests feasible. Full details of the measurement protocol appear in the companion paper (Hernani-Limarino, 2026a); replication data are available at Harvard Dataverse (`https://doi.org/10.7910/DVN/KZMHGU`) and GitHub (`https://github.com/wernerhl/ledgers-bolivia`).

Active labour market programs. Card et al. (2018) provide the canonical meta-analysis: average impacts are close to zero in the short run, becoming more positive two to three years after completion. Benhassine et al. (2018) find that inducing Benin's informal firms to formalize produced zero profit gains. The present paper provides a micro-behavioral explanation for these modest effects: the majority of loss-making informal workers are not rationing-constrained (Type D) but voluntarily trapped (Type C). Programs that raise the formal offer arrival rate ρ are irrelevant for workers who would decline such offers. The mistargeting ratio—Type C per Type D at the statutory minimum wage—is 2.96:1.

Contribution. This paper fills five gaps in a mature literature: (1) no structural model of informality incorporates imperfect profit observation; (2) no theory connects profit mismeasurement to firm persistence; (3) the marginal-versus-average return reconciliation that explains how high experimental returns coexist with 43 percent loss-making rates is unstated; (4) no micro-behavioral theory of ALMP mistargeting exists; (5) existing measurement focuses on survey-diary gaps, but no paper provides a transaction-level decomposition of the accounting gap into its components.

Organisation. Section 2 develops the model. Section 3 derives the five empirical predictions. Section 4 describes the diary instrument and data construction. Section 5 presents the empirical tests. Section 6 develops the policy implications. Section 7 proposes an improved measurement instrument for future data collection. Section 8 concludes.

2 Model

2.1 Environment

Time is discrete and infinite. A unit measure of agents is indexed by innate ability $a \sim F(a)$, drawn once at birth and fixed. Each period every agent occupies one of three states: informal self-employment (I), formal wage employment (W), or unemployment (U). The model is general: it applies to any economy in which informal self-employment is a viable alternative to formal employment and agents accumulate social infrastructure through operation.

Assets. Each agent holds wealth $A \geq 0$. Informal agents face a binding borrowing constraint: no formal credit exists for agents without collateral records, so working capital satisfies $k \leq A$ in every period.

Formal labour market. A fixed measure of formal wage positions exists, each paying wage w^* per period. Each period, any unmatched agent receives a formal offer with probability $\rho \in (0,1)$—reflecting both the scarcity of formal positions and matching frictions. A matched formal worker separates exogenously with probability $\lambda \in (0,1)$ per period.

Idiosyncratic shock. Each period of informal operation the agent draws $\varepsilon \sim G(\varepsilon)$ with $E[\varepsilon] = 1$, $\text{Var}(\varepsilon) = \sigma_\varepsilon^2 > 0$, support on $\mathbb{R}_+$, independently across agents and periods. The shock captures transitory variation in demand, input quality, and operating conditions that is outside the agent's control.

2.2 Dual Embedded Capital

Two stocks of embedded capital accumulate through informal operation and depreciate upon exit.

Productive capital e^P. Labour-augmenting knowledge: operational routines, technical skill, production efficiency. It raises the marginal product of hired labour and therefore raises both output and economic income. It is portable: the agent carries it regardless of where she operates.

Relational capital e^X. The accumulated social infrastructure of informal operation: loyal customer relationships built on years of personal familiarity; supplier credit extended on the basis of demonstrated reliability; landlord or municipality relationships that convert fixed cash obligations into state-contingent deferrals; guild membership and location rights earned through tenure. Relational capital is not portable. The loyal customer base disperses when the operation moves. The supplier credit evaporates when the relationship ends. The landlord's goodwill is place- and person-specific. Relational capital does not enter the production function: it generates no additional output per unit of labour. Its effects are entirely on the cash side.

Common law of motion. Both components follow an identical law of motion:

$$e^{j\prime} = (1 - \delta_e)\, e^j + \varphi^j \cdot \mathbf{1}[O = I], \qquad j \in \{P, X\}, \tag{1}$$

where $\delta_e \in (0, 1)$ is the common depreciation rate and $\varphi^j > 0$ is the flow accumulation rate during informal operation. Upon exit, both components depreciate with no new accumulation.

The common law of motion (1) generates the fundamental identification problem that motivates the paper's measurement strategy. Both e^P and e^X are monotone increasing functions of informal tenure $T_i = \sum_t \mathbf{1}[O_{it} = I]$. No external observer can separate them from tenure alone: a ten-year vendor has accumulated both productive knowledge *and* a loyal customer base and a landlord relationship, but no survey variable records which component is larger. This is not a measurement failure that better surveys can correct in the usual sense—it is a consequence of the economics of informal capital accumulation. What separates e^P from e^X is not the time spent in operation but what was built during that time: which component grew faster depends on the agent's sector, location, and social endowments in ways that standard survey questions do not observe. The financial diary breaks this indeterminacy by observing the cash flow consequence of e^X directly, rather than inferring it from tenure.

2.3 Technology and Payoffs

Production. Output under informal operation is:

$$y^I = \varepsilon \cdot a \cdot k^\alpha \cdot (e^P \ell)^{1-\alpha}, \qquad \alpha \in (0,1), \tag{2}$$

where $k \leq A$ is working capital and ℓ is hired labour. Relational capital e^X is absent from (2).

Supplier credit and the fair markup. Supplier credit—a key component of e^X— extends the effective working capital beyond A, potentially allowing e^X to enter production indirectly through an expanded capital stock. The following assumption rules out any net effect on economic income from this channel.

Assumption 1 (Fair Markup)**.** Supplier credit embedded in e^X carries an implicit markup equal to the informal borrowing rate r^{inf}. The productive gain from expanded working capital is exactly offset by the implicit cost, leaving the net effect of supplier credit on adjusted net income equal to zero.

Assumption 1 is a first-order approximation. It is conservative: if the markup is below the fair rate —as in mature long-tenure relationships—then e^X has a small positive net effect on π^{adj}, making the Type C population smaller than estimated. All results stated below hold under this assumption.[1]

Cash flow. Let c denote cash operating expenses: the actual cash paid for rent, utilities, and inputs in the current period, which differs from the economic cost $\bar{c}$ when relational capital defers or reduces cash obligations. Operating cash flow is:

$$CF = \varepsilon \cdot a \cdot k^{\alpha} \cdot (e^P \ell)^{1-\alpha} - w\ell - c + g(e^X), \tag{3}$$

where $g : \mathbb{R}_+ \to \mathbb{R}_+$ is the net cash contribution of relational capital, with $g'(e^X) > 0$ and $g(0) = 0$. The function $g(e^X) = s(e^X) - \tilde{c}(e^X)$ combines two sub-channels: $s'(e^X) > 0$ captures additional cash inflows from loyal customers and guild reciprocity; $\tilde{c}'(e^X) < 0$ captures cash cost reductions from deferred rent and below-market supplier arrangements, net of their implicit costs under Assumption 1.

[1] The robustness of Assumption 1 is assessed empirically. If supplier credit carries a rate below the informal market rate r^{inf}, the weekly benefit to π^{adj} equals $(r^{inf} - r^{sup}) \times$ payables. In the diary sample, the median payables balance for Type C firms is Bs 49. At a generous informal rate of 10 percent per month (Bs 2.3/week per 100 Bs of payables), the maximum weekly benefit to π^{adj} is Bs 1.1 at the median—against a median $|\pi^{\text{adj}}| =$ Bs 354. Even if all supplier credit were entirely free, only 2 of 150 Type C firms (1.3 percent) would be reclassified as viable. Assumption 1 is therefore not binding: the payables stock is too small relative to the income gap to affect the typology. This finding is consistent with the observation that relational capital in this sample operates primarily through cash cost reductions (deferred rent, mutual aid) rather than through large credit balances.

Because ε is stochastic, CF is a random variable with first two moments:

$$E[CF] = ak^{\alpha}(e^{P}\ell)^{1-\alpha} - w\ell - c + g(e^{X}), \tag{4}$$

$$\sigma_{CF}^{2} = \sigma_{\varepsilon}^{2} \cdot \left[ak^{\alpha}(e^{P}\ell)^{1-\alpha}\right]^{2} \cdot h(e^{X}), \tag{5}$$

where $h : \mathbb{R}_{+} \to (0,1]$ satisfies $h(0) = 1$ and $h'(e^{X}) < 0$. When $e^{X} = 0$, the full variance of ε passes through to CF. As e^{X} accumulates, relational buffers absorb an increasing share of each shock: the compassionate landlord converts a fixed rent payment into a deferred payment contingent on realised conditions, truncating the left tail of the cash flow distribution.

Adjusted net income. The welfare-relevant income measure charges all costs at economic value, including non-cash items:

$$\pi^{\text{adj}} = E[y^{I}] - w\ell - \bar{c} - wh^{own} - d - r^{inf}b, \tag{6}$$

where $\bar{c}$ is the economic cost of operating space and inputs (market rental value regardless of what is paid in cash), wh^{own} is the opportunity cost of owner hours at shadow wage w, d is economic depreciation, and $r^{inf}b$ is informal debt service. Under Assumption 1, e^{X} is absent from (6).

The accounting identity. Let $\pi^{\text{conv}} = y^{I} - w\ell - c$ be conventional profit—what surveys measure. Define the accounting gap $\Delta \equiv \pi^{\text{conv}} - \pi^{\text{adj}}$. Combining (3)–(6), and using $g(e^{X}) = \bar{c} - c$ under Assumption 1:

$$\Delta = wh^{own} + d. \tag{7}$$

The gap between conventional and adjusted income is driven entirely by two non-cash items: imputed owner labour and economic depreciation. Debt service $r^{inf}b$ does not appear in Δ because it enters π^{conv} directly—it is subtracted before the conventional income measure is formed— and therefore cancels.[2]

The master accounting identity follows immediately. Writing $\eta_i = (\varepsilon_i - 1) \cdot y_i^I$ for the within-period deviation of output from its expectation ($E[\eta_i] = 0$):

$$CF_{obs} = \pi^{\text{adj}} + wh^{own} + d + g(e^X) + \eta_i. \tag{8}$$

Identity (8) holds for every firm in every period. It has four distinct components: π^{adj} is economic income; $wh^{own} + d$ is the non-cash cost gap; $g(e^X)$ is the relational cash contribution, persistent across periods for a given firm; and η_i is the transitory shock, mean zero across firms. In a single diary wave, $g(e^X)$ and η_i are observationally confounded—both appear as deviations of CF from $\pi^{\text{adj}} + \Delta$. With repeated waves they are separately identified: $g(e^X)$ is the within-firm persistent component and η_i is the transitory residual. Section 7 develops the multi-wave measurement design that makes this identification feasible.

Proxy bias in the empirical test. In a single diary wave, the observable proxy for CF is conventional profit π^{conv}, which equals $CF_{obs} - g(e^X) - \eta_i$. Since $g(e^X) \geq 0$ by construction, π^{conv} understates true cash flow by $g(e^X)$ in expectation. This creates a one-sided classification error in the empirical implementation of condition (T*). If $\pi^{\text{conv}} \geq w^*$, then $CF_{obs} \geq w^*$ almost surely (since $g(e^X) \geq 0$): no false positives. If $\pi^{\text{conv}} < w^*$, then CF_{obs} may still exceed w^* if $g(e^X) > w^* - \pi^{\text{conv}}$: some true Type C firms are misclassified as Type D. Therefore the empirical Type C share computed

[2]This identity is verified exactly in the empirical data: $(\pi^{\text{conv}} - \pi^{\text{adj}}) - (wh^{own} + d) = 0$ for all $n = 500$ diary firms.

from $\pi^{\text{conv}} \geq w^*$ is a *lower bound* on the true Type C share. Every result stated below is conservative in this direction: the true trapped population is weakly larger than the measured one.

2.4 Bellman Equations and the Trap Condition

Preferences. Agents maximise expected discounted utility with discount factor $\beta \in (0,1)$ and utility satisfying $u' > 0$, $u'' < 0$, with coefficient of relative risk aversion $\gamma > 0$. Working with the mean-variance approximation $u(CF) \approx E[CF] - \frac{\gamma}{2}\sigma^2_{CF}$ defines the certainty equivalent:

$$CE(CF) \equiv E[CF](e^P, e^X) - \tfrac{\gamma}{2}\sigma^2_{CF}(e^X). \tag{9}$$

This is exact under CARA utility with normally distributed ε and a first-order approximation otherwise. Under risk neutrality ($\gamma = 0$), $CE(CF)$ collapses to $E[CF]$ and all results below hold as a special case.

Value functions. The three Bellman equations are:

$$V^I = CE(CF) + \beta\big[\rho \max(V^W, V^I) + (1-\rho)V^I\big]\,, \tag{B-I}$$

$$V^W = w^* + \beta\big[(1-\lambda)V^W + \lambda\big(\rho \max(V^W, V^I) + (1-\rho)V^I\big)\big]\,, \tag{B-W}$$

$$V^U = \beta\big[\rho \max(V^W, V^I) + (1-\rho)V^I\big]\,. \tag{B-U}$$

Unemployment income is set to zero ($b = 0$), reflecting the absence of formal unemployment insurance in environments where the informal sector is large.[3] Note

[3]The result generalises to $b > 0$ provided $b < w^*$, which holds in all calibrated cases. The full expression for V^W under $b > 0$ is given in the appendix.

$V^U < V^I$ whenever $CE(CF) > 0$: unemployment is dominated by informal operation for any active firm, so all transitions out of I go to W, never to U voluntarily.

Derivation of the trap condition. We seek the condition under which $V^I \geq V^W$, so the agent declines a formal offer. Conjecture $V^I \geq V^W$. Under this conjecture, $\max(V^W, V^I) = V^I$ in both (B-I) and (B-W). From (B-I):

$$V^I = CE(CF) + \beta V^I \quad \Rightarrow \quad V^I = \frac{CE(CF)}{1-\beta}. \tag{S-I}$$

From (B-W):

$$V^W\big[1 - \beta(1-\lambda)\big] = w^* + \beta\lambda V^I \quad \Rightarrow \quad V^W = \frac{w^* + \beta\lambda V^I}{1-\beta(1-\lambda)}. \tag{S-W}$$

Substituting (S-I) into (S-W) and imposing $V^I \geq V^W$:

$$\frac{CE(CF)}{1-\beta} \geq \frac{w^* + \frac{\beta\lambda\, CE(CF)}{1-\beta}}{1-\beta(1-\lambda)}.$$

Multiplying through by $(1-\beta)\big[1-\beta(1-\lambda)\big] > 0$ and collecting:

$$CE(CF)\,\underbrace{\big[1-\beta(1-\lambda)-\beta\lambda\big]}_{=\,1-\beta} \;\geq\; (1-\beta)w^*.$$

Dividing by $(1-\beta) > 0$ yields the result.

Proposition 1 (Trap Condition)**.** *An agent in informal self-employment declines a formal offer if and only if*

$$CE(CF) \;\equiv\; E[CF](e^P, e^X) - \tfrac{\gamma}{2}\sigma^2_{CF}(e^X) \;\geq\; w^*. \tag{T*}$$

The condition is independent of ρ and λ.

Economic interpretation. The cancellation of λ has a clean logic. When $V^I \geq V^W$, the agent prefers I in every period. Every separation from W returns her to I, where she stays. How often she passes through W—the value of λ—is therefore irrelevant: each passage is a detour that ends at the same destination. The frequency of the detour does not change the destination. Job insecurity in the formal sector does not push agents into the informal sector; it merely determines the path of agents who were going there anyway.

The cancellation of ρ is the policy result. An agent who declines every formal offer receives no benefit from more of them. Raising ρ—the canonical active labour market intervention—changes the welfare of Type D agents, for whom each additional offer is a chance at the preferred state. For Type C agents, each additional offer is declined. The programme cost is incurred; the occupational outcome is unchanged.

Corollary 1 (Policy Non-Neutrality). *For agents satisfying* (T*)*, $\partial V^I/\partial \rho = 0$. Raising the formal offer arrival rate has no effect on the occupational choice of trapped agents.*

Proof. Under (T*), $\max(V^W, V^I) = V^I$ regardless of ρ. Then $V^I = CE(CF)/(1-\beta)$, which does not contain ρ. □

Proposition 2 (Dual Role of Relational Capital). *$CE(CF)$ is strictly increasing in e^X through two channels:*

$$\frac{\partial\, CE(CF)}{\partial\, e^X} = \underbrace{g'(e^X)}_{>0} + \underbrace{\left(-\tfrac{\gamma}{2}\right) \sigma_\varepsilon^2 [y^I]^2\, h'(e^X)}_{>0\ for\ \gamma>0} > 0. \tag{10}$$

The level channel ($g' > 0$) and the variance channel ($-h' > 0$) both deepen the trap.

For any $\gamma > 0$, the variance channel is strictly positive, so the trap is strictly tighter under risk aversion than under risk neutrality.

Proof. Differentiate (9) using (4) and (5), with $g' > 0$ and $h' < 0$. $\square$

Proposition 2 identifies e^X as uniquely powerful compared with e^P. Productive capital also deepens the trap by raising $E[CF]$ through higher y^I, but it simultaneously raises the scale of output exposed to ε, making the variance effect of e^P ambiguous in sign. Relational capital faces no such ambiguity: $h'(e^X) < 0$ regardless of the level of ε-exposure. An agent who accumulates relational capital gets both a higher mean and a lower variance. For a risk-averse agent comparing a risky informal stream against a certain formal wage, this double benefit is strictly more valuable than what productive capital alone can provide.

Proposition 3 (Viability Threshold)**.** *There exists a unique ability threshold $a^*(A, e^P, \bar{c})$ such that*

$$\pi^{\text{adj}} \gtreqless 0 \iff a \gtreqless a^*(A, e^P, \bar{c}). \tag{V}$$

The threshold satisfies $\partial a^*/\partial A < 0$, $\partial a^*/\partial e^P < 0$, and $\partial a^*/\partial \bar{c} > 0$.

Proof. π^{adj} is continuous and strictly increasing in a, with $\pi^{\text{adj}} \to -\infty$ as $a \to 0$ and $\pi^{\text{adj}} \to \infty$ as $a \to \infty$. The unique root defines a^*. Comparative statics from the implicit function theorem applied to $\pi^{\text{adj}}(a^*) = 0$. $\square$

Conditions (T*) and (V) are structurally independent. (V) depends on $(a, A, e^P, \bar{c})$ and is absent from the Bellman equations. (T*) depends on $(e^P, e^X, \gamma, \sigma_\varepsilon^2)$ and is absent from the profit function. Their independence is what generates a non-trivial four-type partition: if the two conditions were correlated by construction, the typology would degenerate to fewer populated cells.

Corollary 2 (Equilibrium Typology). *Crossing conditions* (V) *and* (T*) *partitions informal self-employed into four types:*

	$CE(CF) \geq w^*$	$CE(CF) < w^*$
$\pi^{\text{adj}} > 0$	***Type A*** *voluntary informal*	***Type B*** *viable, exposed*
$\pi^{\text{adj}} \leq 0$	***Type C*** *DU, self-trapped*	***Type D*** *DU, rationing-constrained*

Type A and Type C satisfy (T*) *and decline formal offers; raising ρ has no effect on them (Corollary 1). Type B and Type D violate* (T*) *and accept formal offers when available. Standard ALMP is correctly targeted at Type D only.*

The typology is a corollary, not an imposed classification. No agent is assigned a type by observing their outcomes and asking whether they are consistent with the model. The types are derived from the joint sign of two conditions defined on primitives—one from profit maximisation, one from dynamic programming— that are independently derived and structurally unrelated.

Why Type B is empirically thin. Corollary 2 predicts that the Type B cell—viable but exposed—is sparse. A viable firm ($\pi^{\text{adj}} > 0$) has $a > a^*$, implying high $E[y^I]$. By (8), high $E[y^I]$ implies high $E[CF]$ before any contribution from $g(e^X)$. For $CE(CF) < w^*$ to hold alongside $\pi^{\text{adj}} > 0$ requires either a very high formal wage or a risk discount that overwhelms a large positive mean. In the Bolivian context—and in most developing country settings where w^* is near the minimum wage—this crossing is rare. The model therefore predicts Type B is uninhabited at standard parameter values. This is confirmed in the data: Type B contains 2 firms (0.4 percent of the sample) at $w^* \in \{500, 546\}$, rising to 7 firms (1.4 percent) at $w^* = 611$ and 11 firms (2.2 percent) at $w^* = 700$. Type B never exceeds 2.2 percent of the sample across the full w^* range considered. The existence of the Type B cell is a theoretical boundary that keeps the typology honest; its empirical near-emptiness is itself a prediction.

Dynamic persistence. The law of motion (1) implies that the trap tightens endogenously. Every period in I raises both e^P and e^X: $CE(CF)$ increases and a^* falls. An agent who satisfies (T*) today satisfies it more firmly tomorrow. Conversely, every period in W depletes both stocks. An agent who exits Type C to W may re-enter I as Type D if e^X has depreciated sufficiently during the absence—a permanent downward revision in type status. The model therefore predicts hysteresis: exit from the trap is possible but followed by an inferior re-entry point. This is the dynamic rationale for why empirically the trap appears permanent for long-tenure firms: they have accumulated enough e^X that even extended absence from I would not fully deplete it below the threshold.

2.5 Trap Dynamics: Threshold Crossing and Absorption

The results above compare stationary values at a given capital stock. The law of motion (1) makes the model dynamic, and re-introducing the state delivers the paper's sharpest testable implication. Write the informal value as $V^I(e^X)$ and note from (9) that the certainty equivalent is strictly increasing in relational capital: $g'(e^X) > 0$ raises $E[CF]$ while customer and supplier relationships stabilise demand, so $d\sigma^2_{CF}/de^X \leq 0$; both forces push $CE(CF)$ in the same direction.

Proposition 4 (Absorbing Trap)**.** *Suppose $CE(CF)(e^X)$ is continuous and strictly increasing. Then: (i) there exists a unique threshold $\bar{e}^X$ such that condition* (T*) *holds if and only if $e^X \geq \bar{e}^X$; (ii) under continued informal operation, e^X converges monotonically to the steady state $e^X_{ss} = \varphi^X/\delta_e$; if $e^X_{ss} > \bar{e}^X$, every operating firm* *crosses the threshold in finite time; (iii) the trap is absorbing while the firm operates: once $e_t^X \geq \bar{e}^X$, then $e_{t+s}^X \geq \bar{e}^X$ for all $s \geq 0$.*

Proof. (i) follows from monotonicity and continuity of $CE(CF)(\cdot)$. (ii) Equation (1) under $O = I$ is a stable linear difference equation with fixed point φ^X/δ_e; convergence is monotone from any $e_0^X < e_{ss}^X$. (iii) On the region $e^X \geq \bar{e}^X$ the agent declines all offers and remains in I, so (1) continues to apply and $e_{t+1}^X = (1-\delta_e)e_t^X + \varphi^X \geq e_t^X$ whenever $e_t^X \leq e_{ss}^X$. □

Remark 1 (Option value and a lower bound)*.* With accumulation, static condition *(T*)* is sufficient but not necessary for declining an offer. An agent with e_t^X just below $\bar{e}^X$ anticipates that one more period of operation raises e_{t+1}^X, and the continuation value of approaching the threshold can rationalise declining today. The static classification therefore *understates* the behaviourally trapped population: measured Type C is a lower bound.

The life-cycle implication is direct. Entrants hold low e^X, so their certainty equivalent sits below w^*: conditional on being loss-making, young firms are Type D. Operation accumulates e^X; surviving loss-makers migrate from D into C and, by Proposition 4(iii), do not migrate back. Tenure should therefore order the types—$\text{med}(T_D) < \text{med}(T_C)$—and the trapped share of loss-making firms should rise with tenure. Section 5.3 tests both.

3 Empirical Predictions

Six predictions follow from the propositions above. Each is derived; none is imposed. I order them as they will be tested in Section 5: first the accounting foundation (P1), then the central quantitative result (P2), then the observability test (P3), then the mechanism test (P4), then the aggregate implication (P5), and finally the dynamic implication (P6).

Prediction 1 (Non-Cash Origins of Economic Loss)**.** For all loss-making informal firms, operating cash flow is strictly positive. The accounting gap $\Delta = \pi^{\text{conv}} - \pi^{\text{adj}}$ is driven entirely by non-cash costs: imputed owner labour and economic depreciation. No cash changes hands to produce the economic loss; the firm is economically non-viable but financially solvent every operating day.

This follows directly from identity (7). It is not a behavioural prediction but an accounting consistency check. Falsification—a loss-making firm with negative CF—would indicate either measurement error in the diary or genuine financial insolvency, neither of which is consistent with the mechanism. Confirming it validates the measurement instrument and establishes the accounting foundation on which the remaining predictions rest.

Prediction 2 (A Large and Robust Trapped Population)**.** A majority of loss-making informal firms satisfy $CF \geq w^*$ across all defensible values of w^* in the empirically plausible range, and the share is stable as w^* varies within this range.

The model predicts a positive Type C share under any parameterisation in which $g(e^X)$ is non-trivial. It does not pin down the magnitude. The prediction is falsified if the trapped share is near zero, or if it collapses to zero as w^* rises within the plausible range. Stability of the share across w^* values is the evidence that the result is structural rather than a measurement artifact from a particular w^* choice.

Prediction 3 (Observational Indistinguishability)**.** Type C and Type D firms are not significantly different on standard survey variables—age, tenure, owner hours—because both types accumulate embedded capital through the identical law of motion

(1), and tenure predicts both e^P and e^X without distinguishing them. However, Type C and Type D firms are significantly different on asset measures— tool value, initial cash endowment—because e^X accumulation requires physical permanence to collateralise relational arrangements.

The prediction has a methodological implication. No standard household survey separates Type C from Type D, because the variables that separate them require diary-quality asset measurement not present in standard instruments. A programme administrator using survey data to target Type D will systematically contaminate the sample with Type C agents, for whom the programme is irrelevant.

Prediction 4 (Sign Reversal Across Equations)**.** For loss-making firms, the coefficient on family labour hours is negative in the adjusted income equation and positive in the cash flow equation for the same firms. The sign reversal arises because family hours substitute for hired labour: they reduce the cash wage bill (raising CF) while adding imputed labour cost (reducing π^{adj}). The same input moves both measures in opposite directions simultaneously.

This is the paper's most demanding empirical test. It is not a reduced-form correlation. It is a test of whether agents optimise over CF or over π^{adj}. Condition (T*) asserts selection is governed by the certainty equivalent of cash flow. Family hours are exactly the class of inputs that shift the trap condition without shifting the viability condition: they raise $CE(CF)$ relative to w^* while leaving π^{adj} unchanged. An agent who adds family workers is, through the lens of the model, deepening her commitment to Type C—investing in the mechanism that keeps her trapped. If the sign reversal is absent, the model's selection condition has no empirical support.

Prediction 5 (Aggregation Bias in Capital Returns)**.** The pooled capital-revenue elasticity overstates the production-channel return, and the bias is stable across specifications including sector and city fixed effects. The mechanism is a revenue-cash flow divergence: for Type A firms, capital raises revenue and cash flow proportionally; for Type C firms, capital raises revenue but cash flow responds weakly, as the relational structure absorbs the output gain through fixed-price supplier arrangements and loyalty obligations that prevent the revenue increase from flowing through to cash.

For Type A firms, capital enters the production function (2) with coefficient α and the measured revenue elasticity approximates the cash flow elasticity. For Type C firms, the relational structure operates on the cost side—deferred rent, supplier credit, loyalty-based pricing—so capital raises output and revenue without proportionally raising cash flow. The divergence between capital's revenue effect and its cash flow effect is the empirical signature of the relational channel. Under standard treatment evaluation assumptions, a researcher using the pooled revenue elasticity to project the cash flow gain from a capital transfer programme overstates the welfare-relevant return for Type C firms by a factor of $\hat{\alpha}_{pool}/\hat{\alpha}_C^{CF}$, which the data puts at approximately 3.5. The policy implication is the inverse of what high pooled returns suggest: capital does not relax the binding constraint for Type C firms and may deepen the trap by raising $CE(CF)$ through channels other than those measured by the pooled elasticity.

Prediction 6 (The Trap Is Absorbing)**.** Among loss-making firms, the probability of satisfying the trap condition increases with enterprise tenure, and the median tenure of Type D firms is strictly below the median tenure of Type C firms. No firm transitions from C to D while operating.

This follows from Proposition 4: relational capital accumulates under operation, the certainty equivalent is increasing in the stock, and the threshold crossing is one-way. The prediction is distinctive because the leading alternative explanations do not generate it. If loss-making firms declined offers because of a fixed taste for autonomy

(Maloney, 2004), the trapped share would be flat in tenure. If Type C status reflected transitory good cash weeks, it would be uncorrelated with tenure. Only accumulation delivers the gradient. A caveat: in the cross-section the gradient combines accumulation with selective exit—Type D firms accept formal offers and leave the sample, which also raises the trapped share at high tenure. Both channels are implications of the same model; separating them requires panel data.

4 Data

4.1 The Double-Entry Financial Diary

The empirical tests require three objects that are not available in any standard household survey: operating cash flow separate from economic income, asset stocks measured at replacement value, and household composition at the time of operation. We construct all three from a purpose-built seven-day double-entry financial diary administered to 500 informal owner-operated firms across six sectors and nine departments of urban Bolivia in September–October 2024.

The fieldwork window was chosen deliberately. By September the winter school recess (July–August) has concluded and household expenditure patterns have returned to their term-time baseline, restoring demand for gastronomía, personal services, and petty commerce to typical levels. The dry season remains fully established, ensuring that construction—the sector most sensitive to precipitation—operates without weather-induced interruption. The window sits at maximum temporal distance from Bolivia's two principal revenue distortions: the Alasitas–Christmas demand surge (December–January), which inflates commerce and transport revenues by an estimated 40–60 percent above the annual mean, and the Carnival–Semana Santa disruption corridor (February–April), which compresses activity across all sectors. The civic strike calendar is historically quieter in September than in October or November, when departmental autonomy commemorations periodically produce half-day paros in Santa Cruz and Tarija. The combination of stable demand, uninterrupted supply chains, and absence of calendar anomalies means that a seven-day diary fielded in this window yields weekly cash flows that are defensible as representative of the firm's modal operating week—the relevant counterfactual for computing π^{adj} and implementing the typology.

The diary records every transaction at daily frequency using a double-entry structure—25,338 transactions across 500 firms over seven days. Each entry records both the cash or non-cash flow (debit or credit) and the counterpart account (inventory, receivables, payables, equity). This structure enforces the accounting identity $CF_{obs} = \pi^{\mathrm{adj}} + wh^{own} + d + g(e^X) + \eta_i$ at the transaction level, enabling exact decomposition of the income-cash flow gap for every firm. Enumerators visited each firm at the end of each operating day to record the day's transactions together with the owner. Opening and closing cash balances were verified against the transaction sum; discrepancies exceeding Bs 5 triggered a reconciliation round.

4.2 Sample Composition

The sample covers six sectors: Comercio (retail and wholesale trade, $n = 182$), Transporte ($n = 114$), Manufactura ($n = 67$), Gastronomía ($n = 54$), Construcción ($n = 50$), and Servicios Profesionales ($n = 33$). Nine departments are represented, with La Paz (179), Santa Cruz (90), and Cochabamba (84) accounting for 71 percent of the sample. Owners are 56 percent male; median age is 39; median schooling is 12 years (secondary completion). Median enterprise tenure is 7 years, ranging from 1 to

54. Table 1 presents summary statistics for all four types.

Table 1: Summary statistics by firm type

	Full (n=500)	Type A (n=284)	Type C (n=150)	Type D (n=64)
A. Income and assets (medians, Bs/week)				
Weekly revenue	1,802	2,213	1,526	955
Conv. income π^{conv}	1,149	1,586	865	367
Adj. income π^{adj}	140	657	−354	−658
Accounting gap Δ	1,002	859	1,334	1,013
Asset value	31,078	34,923	33,039	15,122
B. Owner and household characteristics				
Enterprise tenure (yrs)	7.0	8.0	7.0	5.0
Owner age (yrs)	39.2	40.1	39.2	37.2
Education (yrs)	12.0	12.0	12.0	12.0
Owner hours/week	48.0	40.0	60.0	49.5
Family hours/week	18.9	5.9	34.9	22.3
Family workers	1.0	1.0	1.0	1.0
Male owner (%)	56	54	62	48
Below poverty line (%)	31	32	27	39
Household size	4.0	4.0	4.0	4.0

Notes: Medians in Panel A and for household characteristics except Male owner and Below poverty line (means). Type B ($n = 2$) omitted. Asset value is replacement cost of tools and equipment. Poverty status and household variables from linked EH household survey. $w^* = 546$ Bs/week. The large accounting gap Δ for Type C relative to Type A (Bs 1,334 vs. Bs 859) reflects the higher owner hours worked by Type C owners (60 vs. 40 hours/week): both types impute owner labour at the same shadow wage, but Type C owners work more hours by choice, deepening the gap between cash flow and adjusted income.

The diary is linked to Bolivia's Encuesta de Hogares (EH) household survey via a folio identifier, providing household composition, per-capita income, and poverty status for each firm's household. Household size averages 4 members; median working-age members (15–65) is 2, of whom a median of 1 is employed outside the firm.

The folio link is verified rather than assumed. At enrolment, enumerators recorded four demographic markers for each firm owner from the diary interview: age, sex, years of schooling, and household size. These were matched against the corresponding EH record to confirm the folio identification. Any discrepancy on two or more markers triggered re-interview to resolve the mismatch; 8 of the 500 records required a second visit and were successfully reconciled. The variables drawn from the EH for this paper— household composition, employment status of household members, and poverty headcount— are household-level characteristics recorded by a different surveyor and not susceptible to the revenue undercount documented in Section 4.4. Revenue undercount is specific to the income question; it does not contaminate household roster data, employment status of other household members, or asset ownership records. The measurement critique applies to EH-based income measures; it does not impugn the household variables used as controls and instruments in Sections 5.4 and 5.5.

4.3 Income Concepts

Four income measures are constructed for each firm, corresponding exactly to the model's accounting decomposition.

Conventional profit π^{conv} equals weekly revenue less cash operating expenses, debt service, and own consumption drawn from the business. It is the measure closest to what an EH surveyor elicits with a single-question income module, and it serves as the observable proxy for operating cash flow in the empirical tests of condition (T*).

Adjusted net income π^{adj} equals π^{conv} less imputed owner labour (owner hours times the shadow wage) and economic depreciation. It is the welfare-relevant measure that determines whether a firm is economically viable. The shadow wage is the national statutory minimum wage divided by 40 hours, equal to Bs 13.64 per hour in 2024, yielding a baseline $w^* = 545.6$ Bs/week.[4] We treat this as the lower bound of

[4]The trap condition (T*) asks whether the agent prefers to remain informal rather than accept a formal job. Formal employment in Bolivia is a fixed-hours contract: the standard workweek is 40

the empirically plausible range for w^* and present all typology results parametrically over the range Bs 500–800.

The accounting identity $\pi^{\text{conv}} - \pi^{\text{adj}} = wh^{own} + d$ is verified exactly at the firm level: across all 500 firms the residual is zero to within Bs 0.01. This verification is non-trivial—it requires that every transaction is correctly classified and that imputed labour and depreciation are computed from the same underlying data as conventional profit. It is the empirical anchor on which all subsequent results rest.

The EH-based income measures—$\pi^{\text{conv},EH}$ and $\pi^{\text{adj},EH}$—are also available for each firm via the household survey link. The EH-based income measures are used exclusively for the poverty and household composition analysis in Section 5; all typology results use diary-based measures.

4.4 What Standard Surveys Miss and Why It Matters

Imagine a household survey enumerator interviewing the taquero from the introduction. She asks: "How much did your business earn last week after paying for supplies, rent, and other costs?" He answers honestly. He reports cash in minus cash out. He does not subtract the market value of the sixty hours he worked, because no cash changed hands for those hours. He does not subtract depreciation on his equipment, because the equipment still works. He reports conventional profit π^{conv}. The enumerator records it, the analyst receives it, and the analysis proceeds as if it were economic income. The entire fifty-year literature on informal sector income is built

hours. The relevant comparison is therefore between the agent's total weekly informal cash flow (at the agent's chosen hours) and the formal weekly wage (at 40 hours fixed). The 40-hour threshold is correct for this comparison. The observation that Type C owners work a median of 60 hours per week (Table 1) does not raise the relevant threshold: the additional 20 hours reflect the agent's optimal choice under informal self-employment and are not available under formal employment. The leisure cost of longer informal hours is a genuine limitation of the current specification, which takes hours as given rather than solving the joint hours-occupation problem. Incorporating a disutility of hours would shift the effective threshold upward, reducing the Type C share at any given w^*. This is noted as a direction for future work.

on this exchange.

The dual-sample design here—the same 500 firms observed by both a diary instrument and the linked household survey— provides a direct, quantified answer to the question of what this exchange loses. The answer is: most of what matters. There are three compounding errors.

Error 1: Revenue undercount. The most familiar problem is that informal operators under-report revenue when asked. They estimate rather than recall, they may deflate for tax reasons even when there is no formal obligation, or they simply forget smaller transactions. The diary, which records every transaction at the end of each operating day, captures 24 percent more revenue than the linked household survey (median diary/EH revenue ratio: 1.24), consistent with the well-documented downward bias in self-reported informal income (de Mel et al., 2008; McKenzie and Woodruff, 2008). The undercount is pervasive—present at every percentile— but its magnitude varies substantially across sectors: the diary exceeds the survey by 1.46 times in Comercio but by less than the survey in Servicios Profesionales (ratio 0.51x), where project-based billing may make monthly recall more reliable than weekly recording. The directional effect is uniform: undercount pushes measured income below the formal wage threshold, incorrectly classifying genuinely voluntary operators as involuntarily trapped.

Error 2: Non-cash costs are invisible to surveys. This is the more important error, and the less discussed one. Even if a survey measured revenue with perfect accuracy, it would still produce the wrong income concept. The reason is that the costs that matter most for identifying loss-making firms—imputed owner labour and economic depreciation—generate no cash outflow and therefore never appear in the survey response.

To make this concrete: the taquero who works sixty hours a week reports his cash expenses without subtracting the value of sixty hours at the market wage, because he never paid himself those wages. The median accounting gap in this sample—the amount a perfect-revenue survey would still get wrong—is Bs 1,002 per week, equivalent to 88 percent of median conventional income. The implication is stark: 208 of the 500 firms in this sample (41.6 percent) have positive conventional income but negative adjusted income. They are economically loss-making—the owner earns below the market wage, capital earns below its opportunity cost—but financially solvent. No household survey can identify them, because the signal of loss-making status lives entirely in the non-cash costs that the survey question cannot reach. These firms are fully invisible to every dataset built on the single-question approach.

Error 3: The two errors do not cancel. A natural question is whether revenue undercount and invisible non-cash costs push in opposite directions and roughly offset. They do push in opposite directions: revenue undercount lowers measured income relative to truth, making operators look poorer than they are; invisible non-cash costs raise measured income relative to adjusted income, making operators look richer than they are. But the offsets are neither equal nor uniform. In this sample, revenue undercount dominates: the survey identifies 351 loss-making firms against the diary's 214, a 64 percent over-count. More importantly, the two errors are sector-heterogeneous—revenue undercount ranges from 0.51x to 1.46x across the six sectors, and the gap-to-income ratio ranges from 61 percent in Manufactura to 109 percent in Gastronomía— so they compound in ways that cannot be corrected by adding a constant to survey income.

The cumulative damage to typology classification is summarised in Table 2 and illustrated in Figure 1. The bottom line is blunt. Of 150 true Type C firms, the survey correctly identifies 84 and sends 63 to the wrong bin. More damagingly, the survey drags 109 true Type A firms—economically viable operators with no policy problem whatsoever—into the category it calls "involuntary informal." False positives (109) outnumber true positives (84). A government programme that targets every survey-identified "involuntary informal" operator would direct 40 percent of its resources to genuinely viable firms that need nothing, 42 percent to trapped firms for which ALMP is irrelevant, and only 18 percent to the rationing-constrained firms that would actually benefit. The problem is not that the survey is slightly noisy. The problem is that it measures the wrong object and produces a targeting list that is, by any standard, worse than random among the firms most relevant for policy.

Table 2: Typology misclassification: survey vs. diary ($n = 500$)

	Survey typology				
Diary typology	Type A	Type B	Type C	Type D	Total
Type A (viable, trap satisfied)	135	9	**109**	31	284
Type B (viable, trap not satisfied)	0	1	0	1	2
Type C (loss, trap satisfied)	3	0	**84**	63	150
Type D (loss, trap not satisfied)	0	1	9	54	64
Total	138	11	202	149	500

Notes: Survey typology uses EH-based $\pi^{\text{adj},EH}$ for viability and EH-based $\pi^{\text{conv},EH}$ for the trap condition, at $w^* = 546$ Bs/week. Diary typology uses diary-based measures (Section 4.3). Bold cells: 109 true Type A firms classified as survey Type C outnumber the 84 true Type C firms correctly identified. Survey precision for Type C identification: $84/(84 + 109 + 9) = 41.4\%$. The survey recovers Type D well in isolation ($54/64 = 84.4\%$ recall) but overwhelms those correctly identified with false positives.

Why cross-country validation requires diary-quality data. The practical upshot for external validity is direct. Cross-country validation of the sector gradient (Table 4) using standard harmonised household survey microdata—SEDLAC, LSMS,

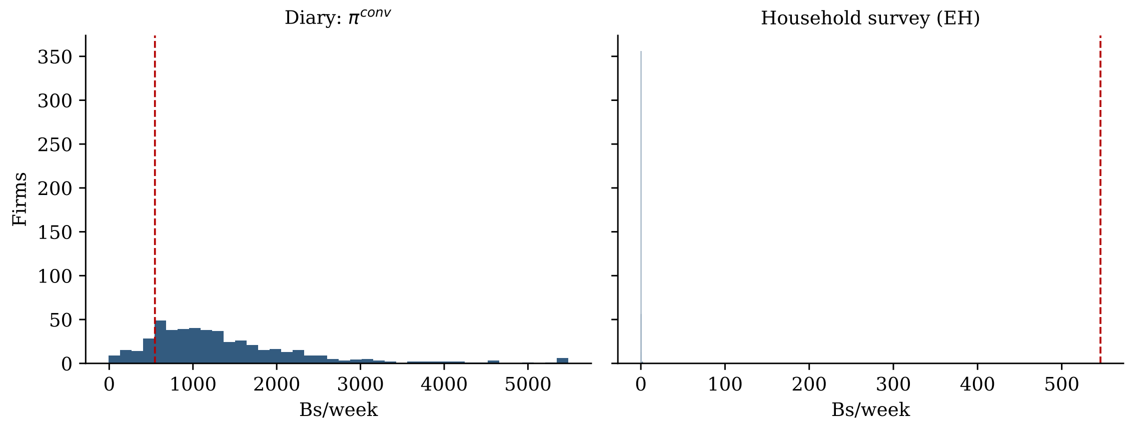

Figure 1: Survey income vs. diary income: the same 500 firms, two instruments

Notes: Left panel: diary-measured conventional income π^{conv}. Right panel: EH household survey income for the same firms. The dashed vertical line is $w^* = 546$ Bs/week. The 208 firms with diary $\pi^{\text{conv}} > 0$ but diary $\pi^{\text{adj}} < 0$— economically loss-making but cash-positive—are recorded in the left panel's positive region and are indistinguishable from viable firms in both panels. The survey over-identifies loss-making firms by 64 percent because revenue undercount shifts the distribution leftward.

or equivalent— is not feasible. The gradient's dependent variable requires two objects that no survey provides: the decomposition of conventional income into adjusted income and cash flow. Without that decomposition, neither "loss-making" nor "trap-satisfying" is correctly identified, and what looks like a cross-country test of the mechanism is actually a test of whether measurement error patterns covary with sector composition across countries. They do, but that is not the same result. Cross-country validation therefore requires deployment of the diary instrument—or an equivalent that separately records cash transactions and non-cash imputations— in a second country. That instrument does not yet exist at scale outside this sample, and we do not claim the result generalises beyond Bolivia until it does. Bolivia is where the instrument was first deployed because Bolivia is where the problem is largest: the most informal economy in Latin America and, by international comparison, in the world.

5 Results

5.1 P2: Non-Cash Origins of Economic Loss

Of the 500 diary firms, 214 (42.8 percent) have $\pi^{\text{adj}} \leq 0$. Every one of the 214 loss-making firms has $\pi^{\text{conv}} > 0$: operating cash flow is strictly positive for all of them. The accounting identity $\pi^{\text{conv}} - \pi^{\text{adj}} = wh^{own} + d$ closes to within Bs 0.01 for every firm in the sample. No cash changes hands to produce the economic loss. Loss-making informal firms are economically non-viable and financially solvent simultaneously.

This result is not trivial. It requires that the double-entry structure correctly apportions every cash flow to the operating account, correctly computes imputed owner labour at the shadow wage rate, and correctly applies straight-line depreciation to the observed asset stock. Each step is verifiable against the transaction record. The exact closure of the accounting identity validates the measurement instrument and establishes the foundation on which Predictions 2–5 rest.

5.2 P1: A Large and Robust Trapped Population

Figure 2 provides the simplest possible summary of the typology result: a scatter plot of π^{adj} against π^{conv} for all 500 firms, coloured by type. The vertical axis separates viable from loss-making firms; the horizontal axis marks the formal wage threshold. The four quadrants correspond to the four types of Corollary 2. Two features are immediately visible. First, the upper-right quadrant (Type A) is full: most firms are viable and trap-satisfying. Second, the lower-right quadrant (Type C) is large and dense: among the 214 loss-making firms, the majority sit to the right of w^*, meaning their cash flow exceeds the formal wage despite negative adjusted income. This is the population the paper is about.

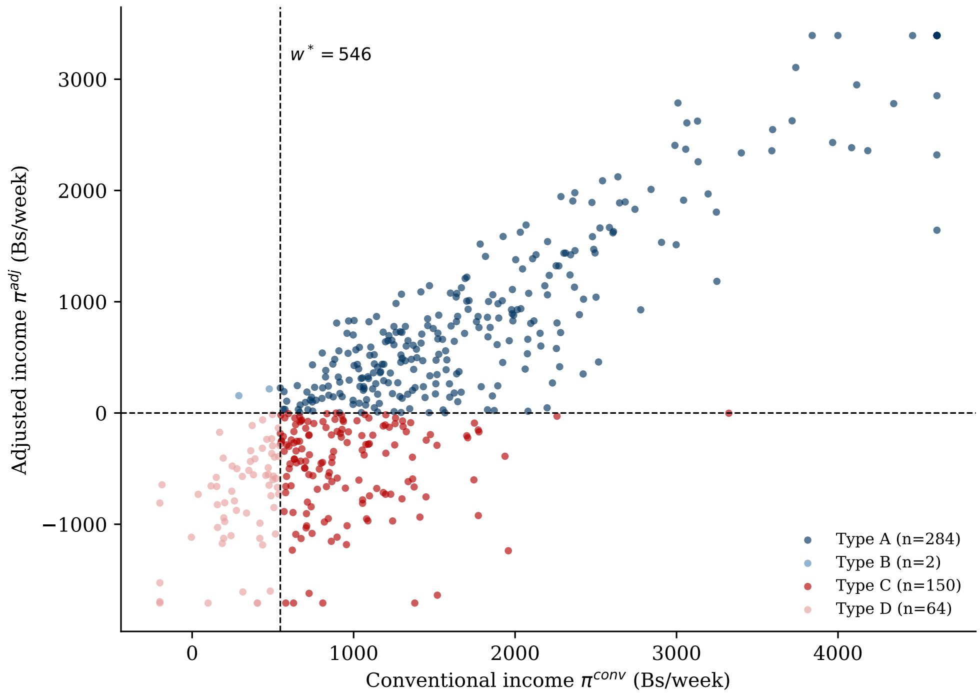

Figure 2: Four-quadrant typology scatter: 500 Bolivian informal firms

Notes: Each point is one firm. Horizontal axis: conventional income π^{conv} (cash flow proxy); vertical axis: adjusted net income π^{adj} (viability measure). Dashed vertical line: $w^* = 546$ Bs/week (statutory minimum wage). Dashed horizontal line: $\pi^{\text{adj}} = 0$ (viability threshold). The lower-right quadrant (Type C: loss-making, trap satisfied) contains 150 firms—70 percent of loss-making firms at this threshold. All 214 loss-making firms have $\pi^{\text{conv}} > 0$, confirming P2 (economic loss is generated entirely by non-cash costs).

Table 3 presents the typology distribution for the full range of w^* values.

Table 3: Equilibrium typology by w^* (n=500)

w^* (Bs/week)	Type A	Type B	Type C	Type D	Loss-makers	C% of loss
500 (statutory min)	284	2	160	54	214	74.8%
546 (shadow × 40h)	284	2	150	64	214	70.1%
611 (upper bound)	279	7	132	82	214	61.7%
700	275	11	107	107	214	50.0%
800	265	21	88	126	214	41.1%

Notes: Condition V: $\pi^{\text{adj}} \equiv$ `net_income_adjusted` > 0. Condition T*: $\pi^{\text{conv}} \equiv$ `net_income_conventional` $\geq w^*$. Both measures are diary-based (see Section 4.3). Type B is theoretically predicted to be near-empty (Section 2.4); the data confirm this at all w^* values. ALMP is correctly targeted at Type D only.

Across the defensible range $w^* \in [500, 611]$ Bolivianos per week—bounded below by the statutory minimum wage and above by the upper-bound shadow wage estimate—between 59 and 75 percent of loss-making informal firms satisfy the trap condition T*. The share is monotone in w^* but does not collapse within the defensible range. Type B is near-empty at all values (0.4–1.4 percent), consistent with the theoretical prediction of Section 2.4. The result first crosses the 50/50 split only at $w^* =$ Bs 700, which corresponds to the shadow wage multiplied by actual owner hours worked—a parameter value that exceeds any standard labour market benchmark for Bolivia in 2024.

The Type C share is a lower bound on the true trapped population for the reason established in Section 2.3: the proxy π^{conv} understates true cash flow by $g(e^X) \geq 0$, so firms with true $CF \geq w^*$ but $\pi^{\text{conv}} < w^*$ are incorrectly assigned to Type D. Accounting for single-week measurement error, the misclassification-robust interval for the Type C share at $w^* = 546$ is $[43.5\%, 86.4\%]$ (Section 5.7). The expected-flip lower bound of 57.4 percent, computed in Section 5.7 from firm-specific weekly volatilities estimated directly on the transaction record, keeps Type C the majority of loss-makers under any redrawn measurement week.

Sector heterogeneity. Table 4 and Figure 3 present the Type C share by sector.

The sector ranking at the baseline threshold is consistent with a prior ordering of relational capital intensity. Transporte has the deepest relational capital: route rights, sindicato membership, location territory, and passenger loyalty are place- and person-specific and irreplaceable upon exit. At the other end, Servicios Profesionales has lower relational capital per Boliviano of capital stock. The Type C share follows this ranking at $w^* = 546$, from 93.6 percent in Transporte to 53.8 percent in Servicios Profesionales. Table 4 shows the gradient is present across the full defensible

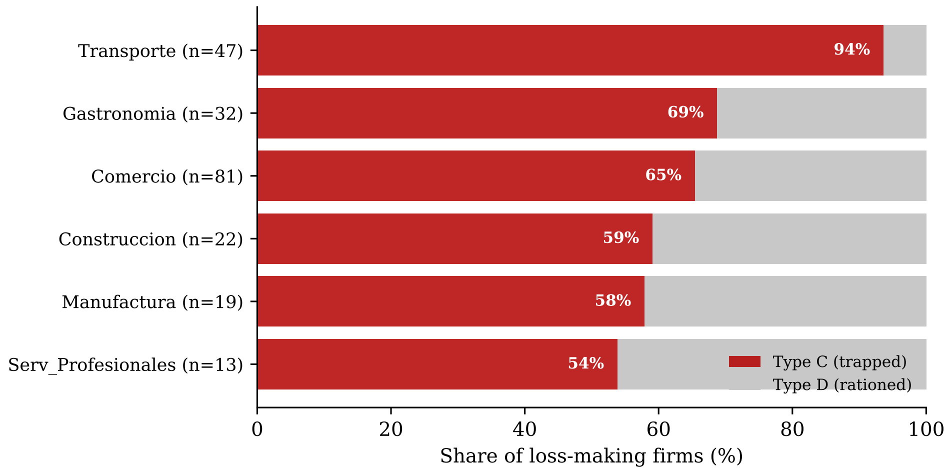

Figure 3: Type C share by sector among loss-making firms

Notes: Horizontal bars show the share of loss-making firms in each sector classified as Type C (voluntary, red) and Type D (rationing-constrained, light). Sectors ordered by theoretical relational capital intensity. The gradient requires no regression: the pattern is either there or it is not. It is there, and it is monotone.

Table 4: Type C share of loss-makers by sector across w^* values

			C% of loss-makers			
Sector	n	Loss	$w^* = 500$	$w^* = 546$	$w^* = 611$	C/D ($w^* = 546$)
Transporte	114	47	95.7%	93.6%	85.1%	14.7x
Gastronomía	54	32	68.8%	68.8%	53.1%	2.2x
Comercio	182	81	71.6%	65.4%	60.5%	1.9x
Manufactura	67	19	63.2%	57.9%	52.6%	1.4x
Construcción	50	22	63.6%	59.1%	40.9%	1.4x
Serv. Profesionales	33	13	69.2%	53.8%	53.8%	1.2x
Full sample	500	214	74.8%	70.1%	61.7%	2.3x

Notes: Sectors ordered by C% at the baseline $w^* = 546$ Bs/week. C/D ratio = Type C / Type D within sector at $w^* = 546$. A chi-square test rejects independence of type distribution from sector at the baseline ($\chi^2 = 39.4$, $p = 0.0006$). Two features are stable across all thresholds in Table 3: Transporte always has the highest Type C share by a large margin, and Type C is the majority type among loss-making firms in every sector at $w^* \leq 546$. The Spearman correlation between the empirical sector C% ranking and the theoretical relational capital intensity ordering is $\rho = 0.94$ ($p = 0.005$) at $w^* = 546$; the ordering is less tight at $w^* = 500$ because the four middle sectors—Gastronomía, Comercio, Manufactura, and Construcción—are theoretically close in relational capital intensity and their exact rank positions vary with w^*. The dominant pattern—Transporte highest, Type C majority everywhere—is robust throughout.

range $w^* \in \{500, 546, 611\}$: Transporte is always first; Type C is the majority type among loss-making firms in every sector at $w^* \leq 546$. This sector gradient is a cross-sectional mechanism test that is independent of the regression evidence in Section 5.5: the model predicts that Type C concentration should co-vary with the intensity of relational capital accumulation by sector, and it does.

5.3 P6: The Trap Is Absorbing—Tenure Dynamics

Proposition 4 predicts that firms age into the trap: entrants are Type D, survivors accumulate relational capital, and the crossing is one-way. Table 5 tests both parts of Prediction 6.

Table 5: Tenure and the trap: testing the absorbing-state prediction

A. Enterprise tenure by type (years)	n	Median	Mean
Type A (viable)	284	8.0	11.0
Type C (trapped)	150	7.0	10.0
Type D (rationed)	64	5.0	8.1
Mann–Whitney, one-sided: C > D, $p = 0.026$; A > D, $p = 0.024$			
B. Trapped share of loss-making firms by tenure			
Tenure bin	n		Type C share (%)
1–3 years	65		61.5
4–7 years	58		72.4
8–15 years	55		72.7
16+ years	36		77.8

Notes: Panel A: tenure distributions by type at $w^* = 546$. Panel B: share of loss-making firms ($n = 214$) classified Type C within each tenure bin; the gradient is monotone. A logit of trapped status on log tenure with sector controls, estimated on the 214 loss-making firms, yields a coefficient of 0.369 (s.e. 0.161, $p = 0.022$); the average marginal effect is 0.069 per log-point of tenure, so moving from the youngest to the oldest tenure quartile raises the trapped probability by roughly 16 percentage points.

Both parts of the prediction hold. The tenure ordering is D < C < A exactly as the accumulation mechanism requires: the median Type D firm has operated five years, the median Type C firm seven, the median Type A firm eight. Among loss-making firms, the trapped share rises monotonically from 61.5 percent in the youngest tenure bin to 77.8 percent among firms operating sixteen years or more, and the tenure gradient survives sector controls. The alternatives are hard-pressed to generate this pattern: a fixed taste for autonomy is flat in tenure, and transitory cash-flow luck is uncorrelated with it. Per the caveat in Section 3, the cross-sectional gradient combines within-firm accumulation with the selective exit of Type D firms to formal employment; both are channels of the same model, and the diary's single wave cannot separate them. What the cross-section does establish is that the data are ordered exactly as an absorbing trap requires and inconsistent with the tenure-neutral alternatives.

5.4 P3: Observational Indistinguishability

Table 6 tests whether standard survey observables separate Type C from Type D among loss-making firms.

Table 6: Separability: Type C vs Type D, loss-making firms

Variable	Type C (n=150)	Type D (n=64)	MW p-value	Separates?
Standard survey observables				
Enterprise tenure (years)	7.0	5.0	0.051	Marginal
Owner age	45.0	41.5	0.375	No
Education (years)	12.0	12.0	0.994	No
Diary-quality measures				
Log tool value	—	—	<0.001	Yes
Owner hours/week	60.0	48.0	0.001	Yes

Notes: Medians reported. Mann-Whitney two-sided p-values. Enterprise tenure: years since current firm founded. Log tool value coefficient from LPM: $t = 6.1$, $p < 0.001$, $R^2 = 0.13$ (assets alone). Adding tenure to this LPM raises R^2 from 0.132 to 0.134.

Age and education do not separate the types. Enterprise tenure is marginally significant at $p = 0.051$ but this is not robust within sectors: in Transporte, where 94 percent of loss-makers are Type C, the tenure difference is 6.5 vs 4.0 years with $p = 0.895$. The tenure difference arises entirely from sectors with heterogeneous type composition.

Owner hours per week strongly separates the types (60 vs 48, $p = 0.001$). This is a new empirical prediction that the model generates without stating it explicitly. Type C owners work longer hours because they optimise over cash flow: additional owner hours reduce the cash wage bill, raising $CE(CF)$ above w^*, at the cost of higher imputed labour, which reduces π^{adj} further. The owner's hours decision mirrors the family hours mechanism tested in Section 5.5.

Log tool value separates the types with $t = 6.1$. This finding is robust: it replicates in the companion double-entry financial diary described in Hernani-Limarino (2026a), where the asset separability result holds with $t = 6.1$ on an independent validation subsample. Asset stock—which requires diary-quality enumeration of replacement values—is the only standard predictor that discriminates between C and D. An LPM for Pr(Type C) including sex, poverty status, age, education, tenure, and log tool value achieves $R^2 = 0.199$; removing all variables except log tool value reduces R^2 to 0.132. Sex is not significant conditional on sector (F-test: $F = 0.674$, $p = 0.412$): the male-female difference in Type C rates is entirely explained by sector sorting.

A companion paper (Hernani-Limarino, 2026b) sharpens this result's boundary: a generalized random forest trained on the diary's ground truth can recover the typology from survey observables with moderate accuracy (Type C $F_1 = 0.68$). The two findings are complementary rather than contradictory. Standard survey variables carry no linear signal that separates C from D—the result above—while a flexible learner *calibrated on diary-quality data* extracts a usable nonlinear one. The diary is not optional in either case: without it there is nothing to train on, and the survey alone misallocates the majority of the loss-making population.

5.5 P4: Sign Reversal and Identification

Table 7 presents the sign reversal test.

Table 7: Sign reversal: family hours and income measures

	OLS		IV (2SLS)	
Outcome	Full (500)	Loss (214)	Full (500)	Loss (214)
π^{adj}	-13.4^{***}	-7.6^{***}	-11.1^{**}	-6.8^{***}
π^{conv} (CF proxy)	$+0.9$	$+5.4^{***}$	$+2.0$	$+5.2^{**}$
First-stage F	12.2	6.3		
Sargan p			0.83	0.39

Notes: Dependent variables: weekly adjusted net income (π^{adj}) and weekly conventional net income (π^{conv}). Endogenous regressor: family hours per week. Instruments: *employed_outside* (household members aged 15–65 employed in jobs outside the firm) and `nn0_15` (children under 15 in household). Controls: log tool value, years operating, sector dummies, city dummies. HC1 robust standard errors. First-stage F for excluded instruments. Sargan-Hansen J-test for overidentification. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

For loss-making firms, the family hours coefficient is -7.6^{***} in the π^{adj} equation and $+5.4^{***}$ in the π^{conv} equation estimated on the same 214 firms. The sign reversal is not a model artefact: it is a consequence of the accounting structure. Family hours substitute for hired labour, reducing the cash wage bill and raising cash flow, while adding imputed labour cost at the shadow wage and reducing adjusted income. The same input moves both measures in opposite directions simultaneously.

Identification. Family hours are endogenous: firms that are more deeply trapped may use more family labour for reasons correlated with the firm's type. We instrument with the number of working-age household members employed outside the firm ($\textit{employed\_outside} \equiv \texttt{em15\_65} - \texttt{n\_family\_workers}$), which measures the supply constraint on family labour imposed by the outside labour market. When more household members hold outside jobs, fewer are available for deployment in the firm. This variation is determined by the household's occupational history, not by the firm's relational capital stock. A second instrument is the count of children under 15 (`nn0_15`), which imposes a caregiving demand on working-age household members that is independent of the firm's type.

The exclusion restriction is supported by the Sargan-Hansen overidentification test: $p = 0.39$ for the π^{conv} equation and $p = 0.83$ for the π^{adj} equation, indicating the two instruments give consistent estimates of the causal effect. Conditional on family hours, neither instrument directly predicts income (all $t < 1$).

The first-stage F-statistic is 12.2 for the full sample and 6.3 for the loss-making subsample. The subsample value falls below the conventional threshold of 10. We note that the OLS and IV estimates are close in magnitude for both outcomes and both samples, indicating that endogeneity bias in the OLS estimates is small. We report both estimators transparently and do not draw causal conclusions that depend on the subsample IV alone. As a further guard, we compute Anderson–Rubin confidence sets, whose coverage is valid regardless of instrument strength, by inverting the joint significance test of the instruments in the structural residual regression. For the loss-making subsample the 95 percent sets are $[-12.8,\ 0.0]$ for the π^{adj} equation and $[-1.0,\ +13.5]$ for the π^{conv} equation: the adjusted-income set excludes all positive effects, the cash-flow set comfortably contains the positive 2SLS point estimate, and the sign reversal is therefore not an artefact of weak instruments.

5.6 P5: Aggregation Bias in Capital Returns

Three results stand out.

Table 8: Capital-revenue and capital-CF elasticities by type

Sample	n	Revenue	CF (π^{conv})	Rev−CF gap	R^2
Pooled	500	0.369***	0.479***	+0.109	0.301
Type A	284	0.289***	0.261***	−0.028	0.277
Type C	150	0.231***	0.106*	−0.124	0.282
Type D	64	0.272***	0.060	−0.212	0.349
Viable (A+B)	286	0.296***	0.269***	−0.027	0.291
Loss (C+D)	214	0.307***	0.381**	+0.074	0.324

Notes: Regressor: log(tool value). Outcome columns: log(weekly diary revenue) and log(π^{conv}). Controls: sector and city fixed effects. HC1 robust standard errors. Bias factor: pooled revenue elasticity / viable revenue elasticity = 0.369/0.296 = 1.25. $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

First, the pooled revenue elasticity (0.369) overstates the production-channel return, proxied by the viable-firm elasticity (0.296), by a factor of 1.25. This bias is stable across all values of w^* in the range 500–800. A researcher using pooled OLS to project the revenue return to a capital transfer programme overstates the production-channel gain by 25 percent.

Second, the revenue-CF divergence differs sharply by type. For Type A firms, capital raises revenue (0.289) and cash flow (0.261) proportionally: the gap is −0.028, essentially zero. For Type C firms, capital raises revenue (0.231) but cash flow responds weakly (0.106): the gap is −0.124, four times larger. This is the empirical signature of the relational channel. Type C firms operate within a social and commercial structure—fixed-price supplier arrangements, loyalty-based customer pricing, deferred rent agreements—that prevents revenue increases from flowing through to cash. Additional capital raises output and revenue but the relational structure absorbs the gain.

Third, the welfare-relevant overstatement is substantially larger than the 25 percent revenue bias suggests. A programme designer using the pooled revenue elasticity (0.369) to project the cash flow gain to a Type C firm overstates the relevant return by a factor of $0.369/0.106 = 3.5$. This is the capital returns analogue of the misallocation ratio: not only does ALMP mis-target the wrong population (Section 5.2), but capital transfer programmes overstate the welfare gain to the targeted population by a factor of three.

5.7 Misclassification Bounds and Risk Aversion Calibration

Misclassification bounds from the transaction record. The single-week measurement introduces noise through the idiosyncratic shock η_i. Rather than proxy within-firm weekly variance with cross-sectional dispersion—which conflates permanent heterogeneity with transitory noise—we estimate it directly from the transaction record. Daily operating cash flow (sales less purchases and operating expenses, excluding imputed entries) aggregates to a weekly total that correlates 0.987 with π^{conv}, validating the construction. Removing pooled day-of-week effects and firm means yields a within-firm daily residual for each of the 500 firms. The residuals exhibit day-to-day mean reversion (median lag-1 autocorrelation -0.195: a strong sales day is followed by restocking), so the implied variance of a weekly total is $\sigma_d^2\,[7 + 2\sum_{k=1}^{6}(7-k)\hat{\phi}^k] = 4.99\,\sigma_d^2$—below the i.i.d. factor of 7. The resulting firm-specific weekly standard deviations have median Bs 335 (interquartile range 248–507).

Two independent checks corroborate this magnitude. First, the validation literature's test–retest reliability of roughly 0.5 for self-employment profits (de Mel et al., 2009) implies a transitory standard deviation of $472 \times \sqrt{0.5} = 334$ Bs/week from our cross-sectional dispersion—within one Boliviano of the transaction-based estimate. Second, the old cross-sectional proxy ($\sigma = 472$) remains available as an ultra-conservative ceiling.

The expected-flip calculation, which weights each firm by its own estimated

Table 9: Misclassification sensitivity at $w^* = 546$

Distance from threshold (Bs)	Type C within band	Type D within band	P(flip) ($\sigma = 335$)
<50	14	13	47.0%
<100	31	22	44.1%
<200	57	35	38.3%
<500	96	57	22.8%
<1,000	139	64	6.8%
Expected flips (firm-specific σ_i): C→D = 27.3, D→C = 15.0			
Type C share: point 70.1%, expected-flip bounds [57.4%, 77.1%]			

Notes: P(flip) evaluates $\Phi(-\,\text{band midpoint}/\sigma)$ at the median transaction-based weekly $\sigma = 335$ Bs. Expected flips use each firm's own σ_i: $\sum_i \Phi(-|\pi_i^{\text{conv}} - w^*|/\sigma_i)$ within type. The lower bound removes all expected C→D flips; the upper bound adds all expected D→C flips. Normal shocks assumed. $n_{\text{loss}} = 214$.

volatility, bounds the Type C share in $[57.4, 77.1]$ percent around the 70.1 point estimate—far tighter than the $[43.5, 86.4]$ interval the cross-sectional proxy implied, and with the lower bound now comfortably above one half. The same machinery applied to the viability margin yields an expected 28 of 214 loss-making firms flipping to viable under a redrawn week, so the 42.8 percent loss-making share is itself robust to weekly noise: at least 37 percent of the sample is loss-making in expectation. Type C dominance is not an artefact of a lucky or unlucky measurement week.

Individual-specific shadow wages. A single threshold $w^* = 546$ charges every owner's hours at the same rate and compares every firm's cash flow to the same formal wage. A 55-year-old owner with six years of schooling and a 30-year-old university graduate do not face the same formal-sector offer. As robustness we construct owner-specific shadow wages from a calibrated Mincer schedule, $\ln w_i^* = r \cdot S_i + 0.03\,X_i - 0.0004\,X_i^2 + 0.10\,\text{male}_i$, with experience X_i defined in the usual way, schooling returns $r \in \{0.06, 0.08\}$ spanning recent Bolivian estimates, and the intercept anchored so the median predicted wage equals the median formal wage of Bs 546. Both the imputed labour charge and the trap threshold then vary across owners (w_i^* runs from Bs 379 at the 10th percentile to Bs 738 at the 90th under $r = 0.06$).

Table 10: Robustness of the typology: wage schedules and depreciation

Specification	Loss-making share	Type C share of loss
Baseline ($w^* = 546$ flat)	42.8%	70.1%
Mincer w_i^*, $r = 0.06$	34.8%	60.9%
Mincer w_i^*, $r = 0.08$	35.6%	59.0%
Depreciation $\times 0.5$	39.0%	67.7%
Depreciation $\times 2$	51.8%	75.3%

Notes: Mincer rows recompute both the imputed owner-labour charge and the firm-specific trap threshold using w_i^*; hourly rates apply w_i^* over a 48-hour formal week. Depreciation rows scale the weekly depreciation charge in π^{adj}. All other accounting components unchanged.

Individual wage schedules lower the loss-making share to roughly 35 percent—high-schooling owners are charged more, low-schooling owners less, and the latter effect dominates at the viability margin—but the trapped share of loss-makers remains 59–61 percent. Halving or doubling depreciation moves the loss share between 39 and 52 percent with the trapped share between 68 and 75 percent. Across every wage schedule and depreciation assumption, Type C remains the majority of loss-making firms.

Risk aversion calibration. At the trap margin, condition (T*) requires:

$$\gamma = \frac{2(E[CF] - w^*)}{\sigma_{CF}^2}.$$

Using π^{conv} as the lower bound for $E[CF]$, the cross-sectional variance of π^{conv} as the upper bound for σ_{CF}^2, and the median Type C firm values, the implied absolute risk aversion is $\hat{\gamma} = 0.00287$ per Bs^2/week^2. Scaling by mean cash flow (Bs 971/week) to obtain relative risk aversion gives $\hat{\gamma} \cdot \bar{CF} = 2.78$, within the standard Arrow-Pratt range of 1–5. The trap is therefore rational at conventional preference parameters.

Since π^{conv} understates $E[CF]$ and cross-sectional variance overstates σ^2_{CF}, the true implied γ is below 2.78: the level channel $g(e^X)$ is doing most of the work, with the variance reduction channel playing a secondary role. Even under risk neutrality ($\gamma = 0$), a firm with $E[CF] > w^*$ satisfies T*.

6 Policy Implications

The paper's policy result follows directly from Corollary 1 and the empirical typology: for agents satisfying condition (T*), the offer arrival rate ρ does not affect occupational choice. Active labour market programmes that raise ρ are correctly targeted at Type D and irrelevant for Type C. The misallocation ratio at the statutory minimum wage—2.96 Type C firms per Type D firm—means that a uniform programme covering all loss-making informal firms reaches its intended beneficiary less than one quarter of the time.

What ALMP can do. Type D agents ($\pi^{\text{adj}} \leq 0$, $CE(CF) < w^*$) satisfy neither viability nor the trap condition. They would accept formal employment if available. Job matching services, employer subsidies, and training programmes that raise labour market competitiveness are welfare-improving for this population. Type D firms are concentrated in Manufactura, Construcción, and Servicios Profesionales, where relational capital is less persistent and the sector-specific Type C share is correspondingly lower. Sectoral targeting of ALMP toward these sectors improves the misallocation ratio substantially.

What reaches Type C. The trap condition $CE(CF) \geq w^*$ is sustained by $g(e^X)$, the cash contribution of relational capital. Three policy instruments can in principle weaken the trap.

First, formalisation incentives that make formal employment more attractive than informal self-employment—raising w^* effectively by providing social insurance, pension accrual, and employment protection—shift the threshold upward and move marginal Type C firms into Type D. The evidence on Bolivian monotributo regimes suggests this channel is weak for established operators with deep e^X stocks: the relational capital loss from exiting informal operation outweighs the incremental formal benefit for firms far above the threshold.

Second, interventions that make the relational infrastructure replicable in the formal sector— portable client records, formal supplier credit at competitive rates, negotiable commercial leases— reduce the e^X depreciation penalty upon exit. This lowers the threshold for voluntary formalisation without requiring the agent to surrender accumulated capital.

Third, sector-level formalisation programmes that coordinate transition—where all operators in a gremio or street market move simultaneously— preserve the relational network while achieving formal status. Individual formalisation destroys the agent's position in the relational structure; collective formalisation preserves it. The sector heterogeneity in Table 4 implies this approach is most promising in Transporte and Gastronomía, where the gremio structure already provides a collective action vehicle.

The targeting problem. The observational indistinguishability result (Section 5.4) means that a programme administrator using EH survey data cannot separate Type C from Type D. The variables that separate them—asset stock at replacement value—require diary-quality enumeration. The policy implication is that accurate typology assignment requires a brief asset enumeration module at enrolment, not the full diary instrument. Five questions are sufficient: replacement cost of primary tool, replacement cost of secondary equipment, current payables balance, current receivables balance, and weekly hours worked. These are observable without a seven-day diary and provide enough information to compute a noisy but useful proxy for e^X.

The welfare cost of the trap. The model generates a welfare number that requires no structural estimation, only the income decomposition that the diary already provides. For a Type C agent in steady state, the weekly welfare cost of remaining trapped is the sum of two components: first, the forgone formal wage—the Bs 546 per week the agent would earn in formal employment but does not; second, the weekly economic loss—the amount by which adjusted income falls short of zero, which the agent implicitly subsidises through uncompensated labour and capital consumption. Together these equal $w^* - \pi^{\text{adj}} > 0$, which is directly observable for each firm.

For the 150 Type C firms in the sample, the median weekly welfare cost is Bs 899— Bs 546 in forgone wages plus a median economic loss of Bs 354. Annualised, this is Bs 46,766 per firm, equivalent to roughly USD 6,800 and 1.65 times the formal annual wage. The trap does not merely represent foregone formal earnings; nearly half the cost is active economic destruction, in the sense that the owner is providing capital and labour at below-market rates while the relational structure prevents realisation of the surplus.

Private rationality and social inefficiency. An objection deserves a direct answer: if the agent's certainty-equivalent cash flow exceeds the formal wage, his consumption possibilities exceed what formal employment offers—in what sense is he worse off? He is not. The trap is privately optimal by construction; the cost is social, and the decomposition of $g(e^X)$ makes its incidence precise. The cash contribution of relational capital is predominantly a bundle of transfers: below-market rent is surplus the landlord forgoes, supplier credit at the fair markup (Assumption 1) is a wash, and loyalty pricing is surplus customers forgo. Transfers redistribute; they do not create. What is destroyed is the allocative margin: the owner's labour produces less inside the firm than it would earn outside it, and $w^* - \pi^{\text{adj}}$ measures exactly that reallocation surplus—the output society recovers if the hours move, gross of the transfers, which simply revert to their counterparties. The Bs 899 median is therefore a social surplus loss borne diffusely by landlords, suppliers, and customers who finance the trap, and by the economy that misallocates the labour—not a private loss to the agent, who is the one party the arrangement serves. To the extent a portion of $g(e^X)$ reflects genuine match surplus rather than transfers (a repeat customer who truly values the specific vendor), the social cost is overstated by that portion; the sectoral gradient in Section 5.2 suggests the transfer component dominates where the trap is deepest.

Table 11 scales these per-firm estimates to the Bolivian urban informal self-employed population using ILO labour force estimates (ILO, 2024) and the sample's Type C share as the prevalence parameter.

Table 11: Aggregate welfare cost of the trap: scale-up scenarios

Assumption	N_C (est.)	Annual cost	% of GDP
Conservative (70% sector coverage, 20% C share)	246,400	USD 1.7B	3.7%
Central (full coverage, 30% C share)	528,000	USD 3.6B	8.0%
Upper (full coverage, 40% C share)	704,000	USD 4.8B	10.7%

Notes: All scenarios use the median per-firm weekly gap (Bs 899); using the mean (Bs 1,025) raises estimates by approximately 15 percent. N_C = implied Type C population in urban informal self-employment. Base population: 1.76 million urban informal self-employed (ILO, 2024). Annual cost = $N_C \times$ Bs $899 \times 52 / 6.86$. GDP denominator: USD 45B (2024, IMF estimate). The conservative scenario restricts coverage to the six sectors in the sample, which account for roughly 70 percent of urban informal own-account employment. This is a static lower bound: the dynamic persistence result of Section 2.4 implies the true welfare cost is higher because the trap tightens over time— accumulated relational capital raises $CE(CF)$ and makes voluntary exit progressively less likely.

These figures should be interpreted as illustrative order-of-magnitude estimates, not structural welfare calculations. They rest on two maintained assumptions: that the sample Type C share (30 percent of informal self-employed) is representative of the broader urban informal population, and that the per-firm welfare gap estimated from a cross-section is a valid proxy for the steady-state cost. Neither assumption is directly testable without data that do not yet exist. What the estimates do establish is that the mechanism is not a curiosity. Even under the most conservative parameterisation, the annual welfare cost of voluntary informal persistence among the economically non-viable is of first-order macroeconomic magnitude for Bolivia— and Bolivia, as the most informally employed economy in Latin America and among the most informal in the world (Schneider, 2018), is the lower bound on the stakes elsewhere.

A final caveat is general-equilibrium. The formal wage w^* is exogenous throughout, which is appropriate for the individual decision problem and for marginal policy but not for the aggregate counterfactual in which a large trapped mass exits simultaneously: half a million workers entering formal labour supply would depress w^*, dampening both the reallocation gain and the incentive to move. The scale-up figures in Table 11 are partial-equilibrium upper bounds on that account, and the policy instruments discussed above operate at the margin, where the partial-equilibrium logic is the relevant one.

7 Measurement Design

The current instrument identifies the trap condition and the typology at a point in time. It cannot separately identify $g(e^X)$ from η_i in a single wave, and it cannot track the law of motion (1) as embedded capital accumulates or depreciates. This section proposes an extended instrument that makes both objects separately identified.

Multi-wave design. Administer four diary waves to the same firms at non-consecutive intervals of four to six weeks. Consecutive waves are uninformative about the law of motion because e^j barely changes between adjacent weeks. Non-consecutive waves expose variation in the trajectory of e^X as the firm ages. The minimum design is Wave 1 (baseline) plus Wave 4 (eight weeks later).

Within each wave, add a daily input module: owner hours (start and end time, break duration), family hours by worker, and main input quantity (kilograms, litres, or units, sector-specific). The daily input module enables estimation of σ_ε^2 from the within-firm variance of output per unit input across days, separately from cross-firm variance. This is the key parameter that current single-wave data cannot identify.

Measuring e^X. Two transaction-level additions separate the level and variance channels of e^X.

Add `customer_id` (an anonymised repeat identifier) to all Module M1A (revenue) entries. This enables computation of the repeat-visit share and revenue concentration index, which proxy for the loyalty component of $g(e^X)$.

Add `payment_status` $\in$ {cash, credit_extended, credit_collected, deferred} to all entries. The share of deferred and credit transactions measures the state-contingency of cash flows that the model associates with $h(e^X) < 1$.

At enrolment and each subsequent wave, administer a five-question Module M8 (relationship inventory): current landlord tenure, supplier with longest credit relationship, number of regular customers recognisable by name, gremio membership, and location stability (months at current site). These questions provide direct proxies for the components of e^X that the transaction record measures indirectly.

Identification of $g(e^X)$ and $h(e^X)$. With four waves, the within-firm mean of $CF_{obs,t} - \pi_t^{\text{conv}}$ estimates $g(e^X) + \bar{\eta}$. The within-firm variance of $CF_{obs,t}$ estimates $\sigma_\varepsilon^2[y^I]^2 h(e^X)$. Two equations, two unknowns, solved separately for each firm. Location change between waves destroys place-embedded e^X (customer base, landlord relationship, location rights) while preserving person-embedded e^X (supplier credit, skills). Ownership change between waves does the reverse. Crossing the two event types provides a within-firm decomposition of e^X into place- and person-embedded components that the cross-sectional data cannot identify.

8 Conclusion

This paper has argued that the standard model of informal sector persistence—agents trapped by formal job scarcity—is wrong for the majority of loss-making informal operators. We develop an alternative model in which informal self-employment generates two forms of embedded capital: productive capital that raises economic income and relational capital that raises cash flow without raising economic income. The model's central result—that an agent declines a formal offer if and only if $CE(CF) \geq w^*$, independently of the formal offer arrival rate and job separation rate—implies that active labour market policy is correctly targeted at a minority of the loss-making informal population and irrelevant for the majority.

Five empirical predictions are tested against a purpose-built double-entry financial diary of 500 Bolivian informal firms. The accounting identity closes exactly at the transaction level. Between 59 and 75 percent of loss-making firms satisfy the trap condition across all defensible formal wage thresholds. Standard survey observables cannot separate trapped from rationing-constrained firms; diary-quality asset measurement can. Family hours raise cash flow and reduce adjusted income simultaneously for loss-making firms, with the sign reversal surviving instrumental variable correction. Pooled capital-revenue elasticities overstate the production-channel return by 25 percent in the aggregate and by a factor of 3.5 for the welfare-relevant cash flow return.

Three results stand out as particularly robust. First, the Type C share varies across sectors in precisely the order that a prior on relational capital intensity predicts, from 93.6 percent in Transporte to 53.8 percent in Servicios Profesionales. This gradient is a mechanism test that requires no regression: the cross-sectional pattern of a theoretically motivated ranking is either there or it is not. It is there.

Second, the trap is rational at standard preference parameters. The implied relative risk aversion is 2.78, within the Arrow-Pratt range, and even under risk neutrality the level channel $E[CF] > w^*$ sustains the trap for the majority of Type C firms. Informal persistence is not irrational. It is the optimal response to accumulated relational capital.

Third, the misallocation is large and measurable. At the statutory minimum wage, 2.96 Type C firms receive each unit of ALMP for every Type D firm that receives it usefully. The ratio falls but remains above unity across the full defensible range of w^*. The policy prescription is not to abandon active labour market policy but to use it where it works, on Type D, and to develop instruments aimed at the relational channel for Type C.

The paper has several limitations. The evidence is from one country and one data collection effort. Cross-country validation of the mechanism is not blocked by theoretical ambiguity but by data availability: Section 4.4 shows that standard harmonised household surveys cannot replicate the typology because the single-question income measure conflates π^{conv} with π^{adj}, misclassifies 42 percent of firms, and produces false positives that outnumber true positives in identifying trapped operators.

Cross-country replication requires the diary instrument in at least one additional country. Whether the same depth of relational capital operates in Peruvian, Ecuadorian, or West African urban informal sectors is an open question that can only be answered with the right instrument. The measurement design proposed in Section 7 would additionally allow direct identification of $g(e^X)$ and $h(e^X)$; the current data identify only their combined effect. The identification of P4 relies on instruments with a weak first stage in the loss-making subsample; the OLS and IV estimates agree in direction and magnitude but the causal interpretation rests on a maintained exclusion restriction rather than a sharp design. These are the constraints of first-wave evidence on a mechanism that, we argue, accounts for a substantial fraction of the informal persistence that development economists have been trying to explain for fifty years—and that has remained hidden because the instruments used to study it could not see it.

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